

A Holistic Design and Simulation of Advanced UTM Services for Urban Air Mobility

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Abstract—This paper presents a modular suite of collaborative Unmanned Aircraft System Traffic Management (UTM) services designed to enhance urban airspace operations. The framework integrates various pre-flight and in-flight services within a semi-autonomous simulation environment that emulates UTM service provision for urban airspace operations. It adopts a holistic approach that fosters intelligent algorithm design, robust serviceability, and strong collaboration among different services, enabling comprehensive simulation and assessment of their automated interactions and overall system performance. Building on the Air Mobility Urban - Large Experimental Demonstrations (AMU-LED) project, this study advances UTM service maturity with over 1,600 simulation flight hours across three traffic density levels within a synthetic urban airspace. The analyses assess the impact of specific UTM services on Key Performance Areas such as safety, path efficiency, and operational workload in a 115 NM² airspace, accommodating up to 150 operations, including 40 simultaneous operations.

I. INTRODUCTION

THE rise of Urban Air Mobility (UAM) is key to the modal development of smart cities, promoting a connected, safer and greener airspace. Global trends are transforming air transportation by integrating Unmanned Aircraft Systems (UAS) with manned aviation. Initiatives such as the Next Generation Air Transportation System (NextGen) in the US and the Digital European Sky (SES2+) in Europe propose a competitive framework aimed at modernising the National Airspace System (NAS) and the SES [1], [2]. This shift includes the creation of UAS Traffic

Management (UTM), which complements traditional Air Traffic Management (ATM) by focusing on low-altitude airspace.

The current development phase of UTM extended by the UAM Concept of Operations (ConOps), aims to challenge innovative operational scenarios to assess potential demand, available resources, and existing capabilities for establishing a functional management ecosystem. These ConOps facilitate UTM service provisions that ensure safe and efficient airspace usage while complying with operational and regulatory requirements [3]. In the US, the Federal Aviation Administration (FAA) collaborates with NASA to develop ConOps and integrate UTM services into the National Airspace System [4]. According to the ConOps [4], the evolution of UAM operations can be divided into initial, midterm, and mature phases, with each phase providing services tailored to operational needs. FAA has also proposed BVLOS rules (Part 108 and Part 146) explicitly recognise Strategic conflict Resolution and Conformance Monitoring as regulated UTM services [5]. In Europe, SESAR JU partnerships initiate projects to promote research and development of UTM services, proposing a Blueprint categorising services into four deployment levels: foundation, initial, advanced, and full (U1 to U4) [6] [7]. The UK Civil Aviation Authority drives UTM technology development through a ten-year roadmap and Future Flight Challenges [8].

As a key element of future UAM operations, UAM services are being developed through both academic research and projects led by government or industry stakeholders. Academic studies typically focus on advanced approaches such as heuristic algorithms, optimization techniques, and machine learning (ML). Research on critical UAM services, include strategic and tactical conflict resolution [9]–[12], risk assessment [13], [14], and demand capacity management [15], [16] have already been widely discussed in the literature. In contrast, research projects usually motivated by practical application. For example, NASA has presents the results of an analysis of strategic conflict management strategies applied to UAM and AAM operations [17], [18], and has also proposed a set of demand capacity balancing algorithms specifically designed for UAM operations [19], aiming to reduce pre-departure delays. The European U-Space programme encompasses a series of influential research projects that propose and develop UAM ConOps, regularly updating UTM service recommendations. While many foundational UTM development projects exist, BUBBLES, Metropolis 2, and DACUS have reshaped Tactical Conflict Detection & Resolution, Strategic Conflict Resolution, and Dynamic Capacity Management services [20]–[22].

To accelerate the integration of UAM operations into the

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airspace and to evaluate the effectiveness of the proposed ConOps and framework, a series of demonstrations have been conducted. In the US, NASA, in partnership with the FAA and industry partners, developed and validated a UTM ecosystem for small UAS through a series of progressively complex field tests [23], [24]. These were structured into four Technical Capability Levels (TCL), each addressing increasing operational complexity – from rural, low-risk operations up to dense urban airspace integration at TCL4. Building on this, NASA launched the National Campaign project to assess and accelerate the technologies required for UAM through a series of systems-level flight demonstrations [17], [25]. In Europe, projects like PODIUM, USIS, SAFIR-Med, and AMU-LED have tested and validated service integration across various scenarios, involving stakeholders such as Common Information Service Providers (CISPs) and UTM Service Providers (USSPs) [26]–[29]. The current focus on deploying U2 services aims to democratise the management of Beyond Visual Line of Sight operations in controlled, low-density airspace where manned aviation is restricted [30]. U-ELCOM implemented UTM services across multiple locations through a new series of trials, testing and demonstrating U1 and U2 solutions [31]. SPATIO focuses on separation management between UAVs, emphasising both Strategic and Tactical Conflict Resolution to maintain safe distances in crowded airspace [32]. EALU-AER is establishing Ireland’s first Digital Sky Demonstrator to integrate UTM technology for enhanced automation through advanced services (U3/U4) [33].

The development of UTM services necessitates extensive testing in synthetic operational environments using simulations. Recent research has highlighted the effectiveness of a comprehensive simulation framework for supporting ATM and UTM services. The TMX traffic manager, created by the Netherlands NLR, has evolved into a medium-fidelity application for exploring innovative ATM concepts [34]. Advances in agent-based techniques have improved the simulation of UTM operational scenarios, ensuring continuous coordination among stakeholders. TU Delft has developed BlueSky [35], an open-source ATC simulator designed to complement projects such as CASSIOPEIA and ELSA, facilitating academic ATM research [36], [37]. NASA’s UTM Laboratory developed Fe3 [38], a versatile simulation component that supports near-term live flight testing and studies high-density, low-altitude air traffic systems, featuring a fast-time simulation option akin to BlueSky. A survey of prominent agent-based frameworks, such as Gazebo, AirSim and Janus, identified essential components for deploying synthetic operations [39]. Cranfield University created a mixed-reality Digital Twin (DT) that accommodates diverse users, including USSPs and ATCs, and assesses operations from various real or virtual platforms [40]. Airbus developed USim that integrates a simulation environment with a strategic conflict resolution prototype [41].

The current UTM state-of-the-art reveals several limitations that hinder the transition from concept to operational deployment. Academic research often address individual functions in isolation, with limited exploration of how services can operate together within an integrated, digitally enabled framework, and with little alignment to governance or industrial practices. This

fragmented approach results in overlapping concepts built on differing assumptions, data dependencies, and no clear path to interoperability. On the industrial side, while notable progress has been made toward deployable UTM systems and their ongoing certifications, these solutions are often proprietary and opaque, offering little awareness into service interactions, performance benchmarking, or operational dependencies. There is a pressing need for integration-oriented research that defines the role of service digitalisation and AI/ML, discusses their place through concrete applications, and clarifies how core UTM services can be jointly implemented, benchmarked under consistent criteria, and designed with transparent inter-service dependencies.

Building on the large-scale AMU-LED project [42], [43], which has applied our research in real-world scenarios, this paper presents additional contributions that further advance the state-of-the-art in UTM service design and simulation.

- 1) A unified UTM service framework is presented that combines an automated pre-flight service toolset with an in-flight service toolset. While the platform supports advanced Digital Twin capabilities (e.g., live-virtual-constructive integration), in this study both toolsets were integrated into a synthetic operational environment that relied exclusively on high-fidelity simulation.
- 2) A modular testbed is developed that facilitates the integration and replacement of specific services, such as alternative models and algorithms. This supports comprehensive simulations that examine interdependencies between services, improving the understanding of how changes in one service impact overall system performance.
- 3) Using the testbed, over 1,600 flight hours of simulation have been conducted to thoroughly evaluate the effects of various UTM services and their parameters, which offers valuable insights into Key Performance Areas, including safety, cost-efficiency, and operational workload.
- 4) EUROCONTROL’s Complexity ANS metrics are adapted to the UTM context, presenting a method for quantifying and managing the complexity of urban airspace operations, which provides benchmarks for assessing how UTM services can alleviate the heightened complexity in dense environments.

The paper is organised as follows. First, a high-level description of the ecosystem embedding the UTM services and their interacting components is provided in Sec.II. The services are grouped into two categories: the pre-flight services, described in Sec.III, and the in-flight services, presented in Sec.IV. This is followed by the experimental evaluation in Sec.V, the discussion in Sec.VI, and the conclusion in Sec.VII.

II. HIGH-LEVEL UTM SIMULATION FRAMEWORK

This section presents a high-level overview of the ecosystem developed in this study. Sec.II-A begins by presenting the overall architecture of the UTM simulation framework. It is followed by how services are orchestrated within this framework in Sec.II-B.

A. Architecture Overview

This research focuses on a set of advanced UTM services designed to be modular toolsets, that is, they can be complemented

with additional services or selectively removed in specific cases. This modularity enables the ecosystem to scale on a service-on-demand basis and constitutes the core interest of this study.

In addition to the UTM services, two additional elements complete the ecosystem. The first is the simulator, which emulates the operational environment in which flight activities are tracked and managed by the UTM services. The simulator can operate in a fully back-end mode or be enhanced with visual components to support situational awareness. In principle, any simulation engine may be used, provided it delivers the required traffic information. Its importance lies in the quality of the generated data, which directly influences the fidelity of the simulated operational dynamics.

Finally, the system accounts for two categories of users: the operator or pilot, who conducts point-to-point flight missions at Very Low Level, and the controller or UTM service provider, who supervises the operation and oversees the execution of UTM services.

Fig.1 illustrates the components described above: the user's mission timeline appears on the left, the simulator is located in the bottom-left corner, and the modular UTM services, which are the main focus of this study, occupy the remainder of the framework. In this research, all operations and services are executed autonomously, without human intervention. The description of the different blocks of the framework are introduced in the following section.

B. UTM Service Provision Flow Summary

As illustrated in Fig.1, the set of advanced UTM services is organised into two main blocks: pre-flight services and in-flight services, which are detailed in Sec.III and Sec.IV respectively. In this paper, we refer to the strategic, pre-flight phase services

that perform the flight authorisation process collectively as pre-flight services, and to the in-flight phase services, often referred to as tactical services, as in-flight services.

In a given piece of airspace managed by UTM services, operations must adhere to a defined protocol, beginning with a flight approval. Pre-flight services process such requests based on existing demand, current operational plans, and the constraints imposed by airspace capacity and risk. At the end of this pre-flight process, the operator typically receives authorization to fly within the airspace along a defined route, during a specified time window. The granted authorization may differ from the original request, and the operation is subsequently registered to enable monitoring throughout its activation.

Once the registered and approved operation enters its authorised time window, it is considered active, and the in-flight services take over the monitoring process. Based on the flight plan and operation tracking, these services assess whether the behaviour of the aircraft aligns with the expected trajectory. If the operation proceeds nominally, no visible intervention is made. However, in the event of an off-nominal situation, the in-flight services are designed to detect the anomaly and respond appropriately to ensure safety.

An intermediate module, referred to as the compiler in Fig.1, acts as a unidirectional bridge between the pre-flight and in-flight services. In the current design, this unidirectional flow implies that in-flight services do not provide feedback to pre-flight components. The compiler processes, organizes, and transmits authorised flight data from the pre-flight services to both the simulator and the in-flight services. While the compiler is also capable of communicating information to third parties (e.g., USSPs, CISP), this functionality falls outside the scope of the present study. Readers interested in large-scale demonstrations of such UTM capabilities are referred to relevant literature [44],

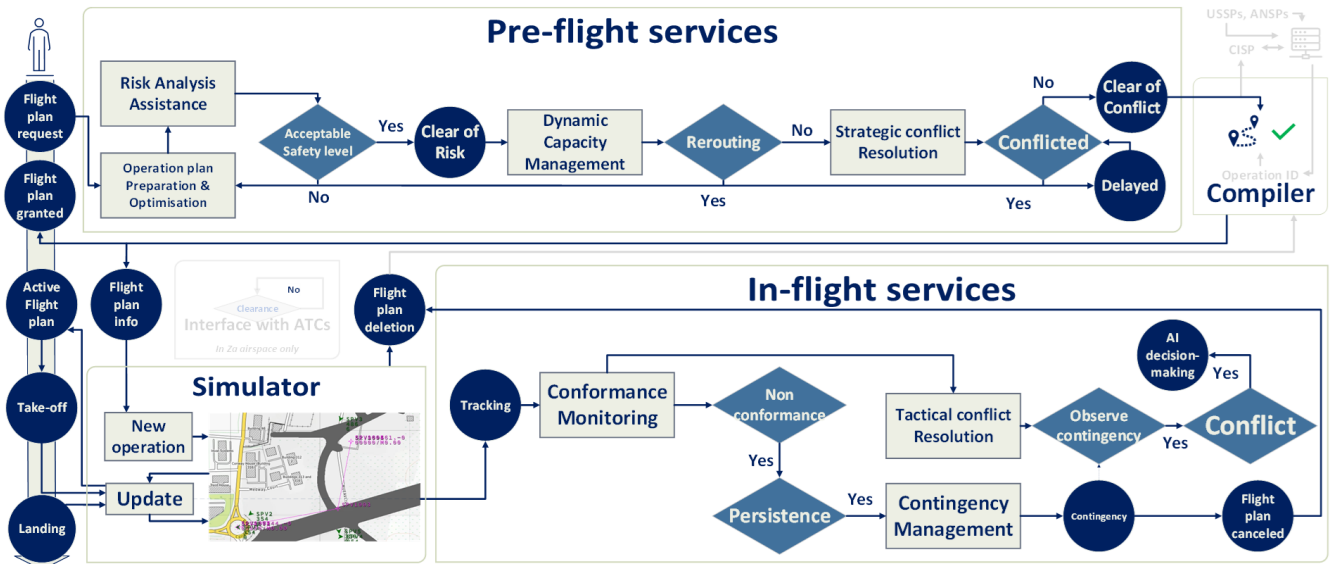


Fig. 1. Overview of the UTM services framework and its interactions. The framework is divided into two stages: pre-flight services, which ensure mission planning before flight authorization, and in-flight services, which oversee execution and respond to operational events in real time.

[45], which discusses findings from the SESAR JU AMU-LED project.

III. PRE-FLIGHT UTM SERVICES

This section focuses on the pre-flight services involved in the flight authorisation process. Sec.III-A provides a high-level overview of their interactions, followed by detailed descriptions of each service: Operation Plan Preparation and Optimisation in Sec.III-B, Risk Analysis Assistance in Sec.III-C, Dynamic Capacity Management in Sec.III-D, and Strategic Conflict Resolution in Sec.III-E.

A. Pre-flight Toolset Integration

The pre-flight service toolset includes the following services: Operation Plan Preparation & Optimisation, Risk Analysis Assistance, Dynamic Capacity Management, and Strategic Conflict Resolution. While these services are presented as black boxes here, they are explained in detail in the rest of this section. The provided references offer additional resources for readers interested in exploring the replicability of each service.

As shown in Fig. 2 and in the pseudo code for the pre-flight phase module through Algorithm 1, pre-flight services are executed sequentially with retroactive effects on the Operation Plan Optimisation service. If an upstream provision encounters a constraint or fails during the flight plan authorisation, these constraints are considered to refine the proposal for a safer and more efficient option. This iterative process continues until all pre-flight services have validated the flight plan for their specific purposes.

The operator submits a flight request with the departure time, origin, and destination, which is processed by the Operation Plan Preparation service (Algorithm 1, line 1). During flight plan construction, static data, including terrain features, ground obstacles, major road networks, and restricted airspace, are considered (line 2). This data represents environmental information that is unlikely to change. The initial flight plan is then sent to the Risk Analysis Assistance service to evaluate its risk level (line 7). Once the flight plan is deemed to exceed the airspace capacity, it is forwarded to the Dynamic Capacity Management service (line 14), which may suggest alternative flight trajectories to avoid congested airspace and balance demand with capacity (line 16). Finally, the flight plan is handled by the Strategic Conflict Resolution service (line 22), which identifies potential conflicts

with other flight plans and resolves them by allocating delays (line 24).

Algorithm 1 Pre-flight Services Workflow

```

1: if operation request (i.e., flight plan =  $\emptyset$ ) then      Sec.III-B
2:   flight plan  $\leftarrow A^*(\text{initial constraints})$ 
3: else
4:   flight plan  $\leftarrow A^*(g_{RAA}, g_{DCM}, g_{ScR})$ 
5: end if
6: for segment in flight plan do                          Sec.III-C
7:   if  $R_{\text{segment}} \geq \text{Risk}_{\text{threshold}}$  then                Eq.1
8:      $g_{RAA} \leftarrow \text{geofence}(\text{segment})$ 
9:   end if
10: end for
11: if  $g_{RAA} \neq \emptyset$  then
12:   return to line 4
13: end if
14: for cell in airspace crossed by flight plan do          Sec.III-D
15:   if cell in congested area then
16:      $g_{DCM} \leftarrow \min(C_r | \text{cell})$                     Eq.2
17:   end if
18: end for
19: if  $g_{DCM} \neq \emptyset$  then
20:   return to line 4
21: end if
22: for cell in airspace crossed by flight plan do          Sec.III-E
23:   if cell is occupied then
24:      $g_{ScR} \leftarrow C_{\text{delay}} | \text{cell}$                     Eq.3
25:   end if
26: end for
27: if  $g_{ScR} \neq \emptyset$  then
28:   return to line 4
29: else
30:   flight plan is approved
31: end if

```

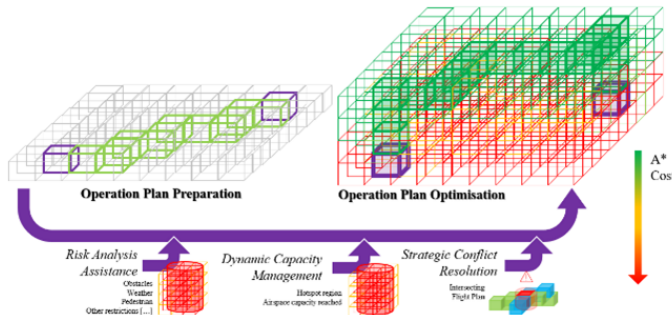


Fig. 2. Pre-Flight services workflow.

B. Operation Plan Preparation & Optimisation

This service includes strategies that optimise the creation and modification of flight plans, ensuring compliance with safety and efficiency requirements for UTM operations.

This service generates initial flight plans based on a discretised 3D airspace divided into equally sized volumes. Each cell connects to its adjacent cells in 10 directions, 8 horizontally and 2 vertically, except where airspace constraints or boundaries exist. Altitude layers are added to these volumes to represent vertical operations, as shown in Fig. 2. Horizontally adjacent cells are dimensioned to maintain the minimum lateral separation required between vehicles, preventing loss of separation events (refer to Sec.IV-D), whereas vertical cell spacing is set in accordance with nominal flight levels and altitude stratification rules defined for the operational concept later in Sec.V-B.

After receiving the original and destination points from flight requests, A* algorithm is used to identify the preferred path through a series of sequential airspace volumes. As detailed in Sec. III-A, static data such as terrain and constraints are considered, including airspace restrictions, significant buildings,

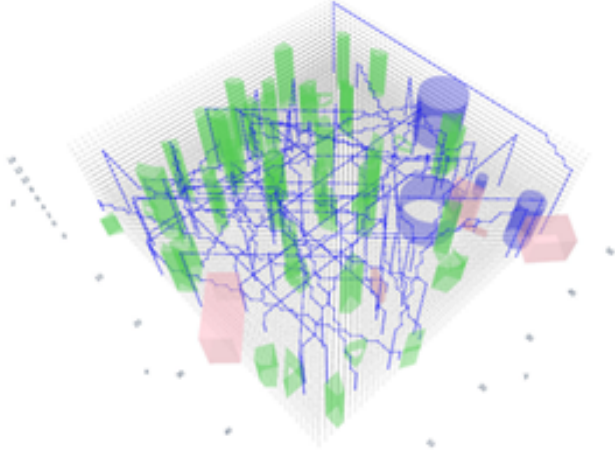


Fig. 3. Operation trajectories generated in the 3D designated airspace with constraints applied [9].

power lines, obstacles and high population density areas. The planning includes three parts: vertical take-off, vertical landing, and the cruise phase. An example of the generated trajectories is illustrated in Fig. 3, where the curved segments represent paths that avoid high-risk or restricted areas.

This service not only generates initial flight plans but also collaborates with other services to optimise trajectories. Through its interface with the Risk Analysis Assistance service (see Sec. III-C), it establishes a feedback loop for risk assessment and trajectory adjustments. If a flight segment's risk level exceeds the safety threshold, geofences based on obstacle circles will indicate high-risk areas. The operation plan optimisation service will then adjust the trajectory accordingly and send the revised routes back to the risk analysis module for reevaluation. This iterative process continues until all flight segments meet acceptable risk levels.

Fig. 3 depicts an optimisation scenario created by the Operation Plan Optimisation service, considering constraints like static obstacles, airspace restrictions, high-risk areas, and hotspot regions, represented by green, blue, and red shapes, respectively. While temporal aspects are not shown, the flight plans in blue are deconflicted, eliminating the need for Tactical Conflict Resolution, as will be discussed later.

C. Risk Analysis Assistance

The Risk Analysis Assistance service conducts both qualitative and quantitative assessments of flight plans to ensure mission area safety. After receiving the flight plan from the previous Operation Plan Preparation & Optimisation service, it segments the route by waypoints and associates each segment with a risk value derived from aircraft performance parameters (e.g., size, weight, cruising speed), environmental factors (e.g., wind speed and direction), and ground context (e.g., population density, road traffic statistics). The module's operational workflow is depicted in Fig. 4, the parameters are collected from operational datasets and live or near-live data feeds.

The risk for each segment is defined as the probability of a fatal event, calculated in Eq. 1, where P_{CR} is the probability of a catastrophic failure during the segment, $P_{IM|CR}$ is the

conditional probability of impacting an object given such a failure, and $P_{FA|IM}$ is the probability of a fatality given the impact. These probabilities are evaluated in accordance with SORA-based safety targets for urban operations.

$$R = P_{CR} * P_{IM|CR} * P_{FA|IM} \quad (1)$$

Once calculated, each segment's risk value is compared to a predefined acceptable threshold. If any segment exceeds the threshold, the system issues a rerouting request to the Operation Plan Optimisation service.

Segments passing over dense urban areas or major roads typically register higher risk values, as seen in the example in Fig.5. Risk severity in the flight plan is visually represented by a colour gradient from green to red, as shown in the same figure. Each segment is colour-coded based on its calculated risk,

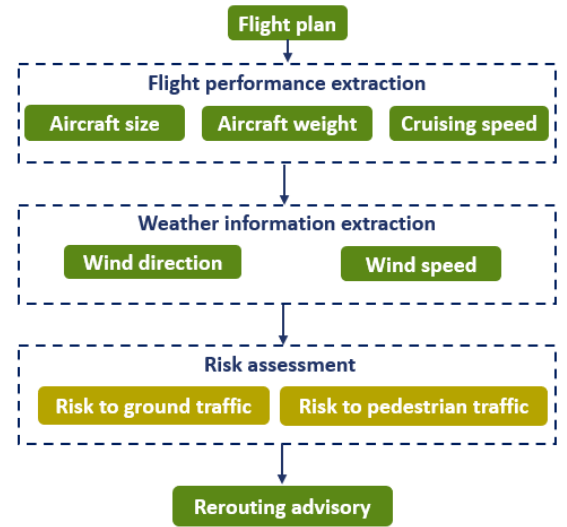


Fig. 4. Risk analysis assistance parametrisation and process [46].

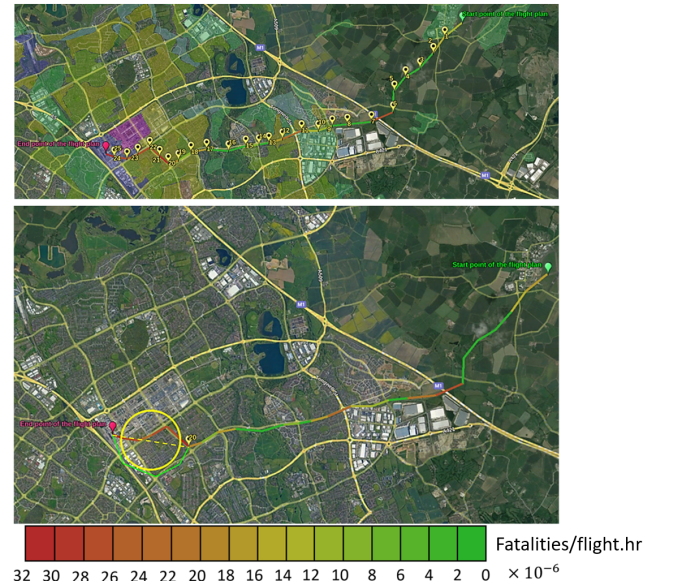


Fig. 5. Risk-based geofencing and local rerouting.

allowing for easy comparison against safety limits. The analysis indicates that segments over highways or densely populated areas show significantly high risks, marked in red. The top figure illustrates risk along the original flight plan. In the same example, the Risk Analysis Assistance identifies a segment with a yellow geofence, leading to an optimised flight plan that includes rerouting constraints for enhanced safety.

This service is also responsible for identifying Safe Landing Zones and Emergency Landing Zones (SLZ & ELZ) using the risk map, selecting clear areas with minimal population exposure where contingent operations can land without compromising surrounding safety. However, it does not perform airspace capacity checks as part of its assessment, as these are the responsibility of the subsequent Dynamic Capacity Management service.

D. Dynamic Capacity Management

The Dynamic Capacity Management service [15] monitors airspace demand based on active flight plans and regulates access when capacity limits are approached. It comprises three threads: capacity, demand, and demand-capacity balancing.

The capacity thread estimates the maximum operations an airspace can support. The airspace is divided into grid cells, where the demand thread assesses traffic demand for each cell using existing flight plans. By integrating the airspace configuration (capacity thread) with the trajectories (demand thread), we can identify intersections between trajectories and airspace. When any airspace portion becomes full and turns into a hotspot, routing is adjusted to avoid it. Alternative trajectories are generated to bypass the affected grid cells, excluding the first and last cells for take-off and landing, by applying these constraints to the Operation Plan Optimisation service (recall Sec. III-B).

Using input from both capacity and demand, a Dynamic Capacity Management model calculates and identifies the optimal trajectory to balance demand and capacity. As shown in Eq. 2, the costs of rerouting operations are quantified.

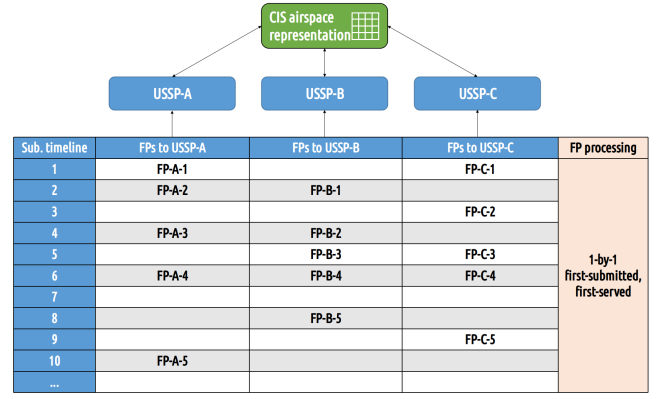
$$C_r = \sum_{f \in \mathcal{F}} \sum_{k \in \mathcal{K}_f} v_f d_f^k z_f^k \quad (2)$$

where d_f^k represents the additional flight duration for alternative trajectory k compared to the originally scheduled duration, v_f indicates the operation's priority based on submission timing, and z_f^k is a Boolean variable that is 1 if trajectory k is selected. Consequently, an optimal rerouting path is identified through the objective function $\min(C_r)$, which mitigates congestion in airspace while minimising the costs.

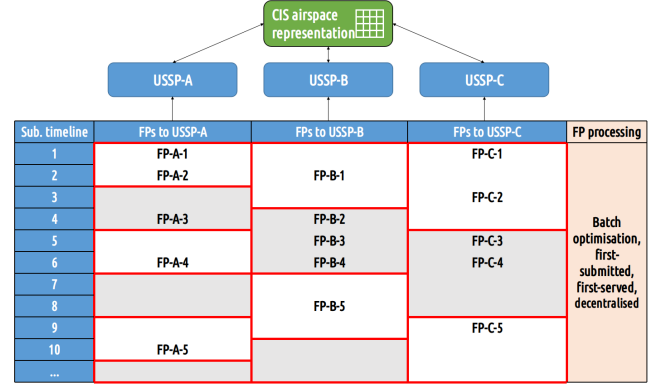
E. Strategic Conflict Resolution

The Strategic Conflict Resolution service aims to identify and address conflicts based on flight plans during the pre-flight phase and after prior service completions. It is designed to greatly prevent the use of the Tactical Conflict Resolution service and the use of airborne collision avoidance systems.

By sharing with relevant parties, flight plans will be compared to the activated ones to check for conflicts. If a conflict arises, a tentative plan change will be proposed, either by delaying



(a) First-Come First-Served (FCFS)



(b) Batch optimisation

Fig. 6. Strategic Conflict Resolution approaches for decentralised USSPs architecture. In (a), each flight plan is processed individually that prioritizes early submissions. In (b), USSPs process multiple flight plans simultaneously using batch optimisation considering spatial efficiency and submission fairness.

the take-off time for specific operations based on a First-Come, First-Served (FCFS) principle or using a batch optimisation approach, as illustrated in Fig. 6a and Fig. 6b respectively. The batch optimisation approach aims to resolve all conflicts while minimising total delay, as shown in Eq. 3.

$$C_{delay} = \sum_{l \in L} \sum_{j \in J_l^{(1)}} \sum_{t \in T_l^{J_l^{(1)}}} (t - r_l^{J_l^{(1)}}) (x_{l,t}^j - x_{l,t-1}^j) \quad (3)$$

The total delay cost is determined by Eq. 3, where $r_l^{J_l^{(1)}}$ represents the initially scheduled take-off times, and t denotes the controlled time of arrival. Only one operation can occupy a cell within a default sliding time window, corresponding to the separation minima. Once the appropriate delay allocation is identified, an iterative negotiation and resubmission process may ensue until all predicted conflicts are resolved. The service provider then assesses airspace occupancy and determines if delays are necessary for any conflicting flights. The FCFS method is illustrated in Algorithm 2.

If a flight plan is conflict-free, it can be activated to update the common airspace representation. This process can occur in parallel, enabling multiple service providers to manage their

Algorithm 2 First-Come First-Served Process.

```
1: for  $f \in F$  do
2:   for  $(e, t), (b_e, t) \in f_i$  do
3:     if  $(e, t)$  or  $(b_e, t)$  is blocked then
4:       delay  $f$  for 1 time unit
5:       go to line 2
6:     else
7:       insert  $(e, t)$  and  $(b_e, t)$  to  $list_l$ 
8:     end if
9:   end for
10:  for  $(e, t)$  and  $(b_e, t) \in list_l$  do
11:    occupancy  $(e, t), (e, t \pm sep)$ 
12:    occupancy  $(b_e, t), (b_e, t \pm sep)$ 
13:    move to the next operation
14:  end for
15: end for
```

flight plans simultaneously. Once a set of grid cells linked to an activated flight is occupied for a specified duration, they will be unavailable for subsequent flight plans until the current operation concludes.

Delay is used as the sole conflict resolution manoeuvre not only because it is the preferred strategy in both academic research [18] and practical applications [47], but also because rerouting has already been applied in the preceding services in the framework, (through both the Risk Analysis Assistance and Dynamic Capacity Management services). Introducing additional geometric modifications during the strategic conflict resolution phase could result in excessive trajectory deviations, potentially altering the operator's original intent for the submitted flight plan.

IV. IN-FLIGHT UTM SERVICES

Sec. III introduced the pre-flight services to deliver flight authorization to the operations. This section is focused on the in-flight services toolset taking over the management of the operations once activated. Sec.IV-A provides a high-level overview of the interactions and workflow of these services, followed by detailed descriptions for each of them: Conformance Monitoring in Sec.IV-B, Contingency Management in Sec.IV-C, and Tactical Conflict Resolution in Sec.IV-D.

A. In-flight Toolset Integration

The in-flight service toolset includes Conformance Monitoring, Contingency Management, and Tactical Conflict Resolution. Each service is detailed in the following sections, with references for those interested in exploring their replicability. Fig. 7 summarises the high-level workflow linking flight status notifications to the activation conditions of in-flight services, which are detailed deeper for each service in the following sections.

At the activation of an operation, Conformance Monitoring defines its 4-dimensional operation volume (OV), specifying both altitude and lateral limits between waypoints. Meanwhile, the Contingency Manager designates safe and emergency landing

Algorithm 3 In-Flight Services Workflow

```
1: Extract operation information from environment
2: for operation in active operations do
3:   Conformance Monitoring is active
4:   Update operation status
5:   Conformance Monitoring ends
6:   Contingency Manager is active
7:   if new operation status is contingent then
8:     operation  $\leftarrow$  Safe Emergency Landing zone
9:     Update Safe Emergency Landing zones capacity
10:  else if contingent operation new status is Ended then
11:    Update Safe Emergency Landing zones capacity
12:  end if
13:  Contingency Manager ends
14:  Tactical Conflict Resolution is active
15:  if operation is conflicted then
16:    operation  $\leftarrow$  resolution manoeuvre
17:  end if
18:  Tactical Conflict Resolution ends
19: end for
```

zones based on a risk map generated by the Risk Analysis Assistance service. In-flight services actively exchange with the operational environment by interpreting traffic information, supplying real-time flight status (see Fig. 7), and issuing instructions, advisories, and resolution commands during off-nominal situations with each function described for its respective service, as shown in Algorithm 3.

At each simulation time step, the in-flight services toolset gathers traffic information from the environment, which the Conformance Monitoring uses to update the current operational status. This involves classifying the operations' statuses continuously. As illustrated in Fig. 7, an active operation can be either nominal or off-nominal. Off-nominal operations may be categorised as non-conformant, contingent, or conflicted. Conformance Monitoring plays a determining role in the collaborative

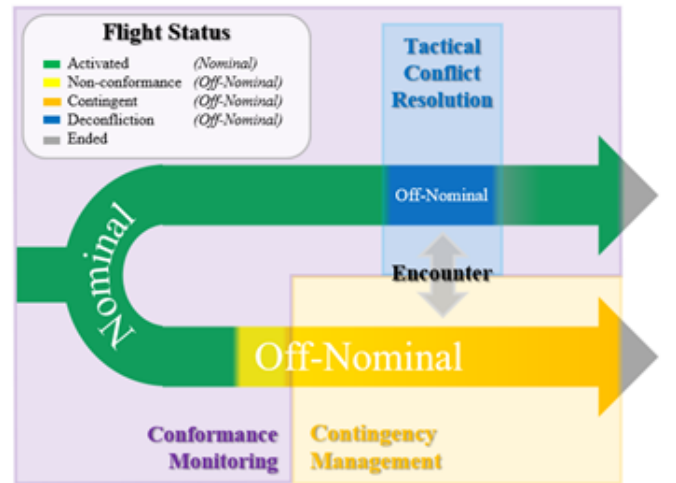


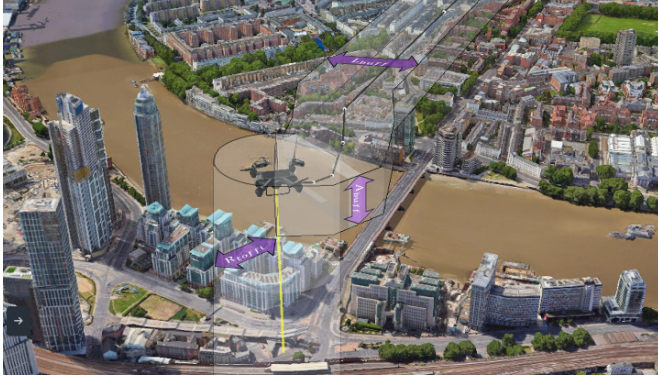
Fig. 7. In-flight services workflow. The activation of the services are driven by the real-time flight status of ongoing operations.

management of operations by delegating tasks to other in-flight services.

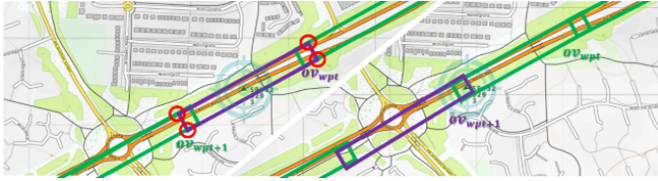
B. Conformance Monitoring

The Conformance Monitoring service [48] ensures continuous verification of an operation's adherence to its designated OV, enabling early detection of deviations and proactive safety management.

The OV, illustrated in Fig.8, defines the permissible 3D space around the nominal flight plan. It consists of two vertical cylinders of radius R_{tofl} extending from ground level to cruise



(a) OV geometry



(b) Intent OV and segment transition shown at two time snapshots

Fig. 8. Illustration of OV geometry and dynamic intent transitions. (a) OV structural components, including take-off/landing cylinders and lateral/vertical buffers defining the permissible cruise corridor. (b) Example of intent-based OV progression at two time snapshots, showing how the active segment transitions from OV_{wpt} to OV_{wpt+1} along the planned route.

altitude at the origin and destination, accommodating manoeuvre tolerances during take-off and landing.

As illustrated in Fig.8a, these cylinders are connected by a polygonal corridor representing the cruise phase, bounded laterally by a buffer L_{buff} and vertically by a buffer A_{buff} that determines the tolerance between two nominal flight levels. The value of L_{buff} reflects the vehicle's manoeuvrability, speed, and traffic priority: a smaller volume increases sensitivity to off-nominal conditions but raises false-alert rates, while a larger volume reduces false alerts but may delay anomaly detection (later explored in Sec.V-D3).

As presented in Fig.8b, the OV is divided into segments based on the nominal waypoints of the flight plan, with each segment representing an intended OV. Let $\mathcal{P}$ be a permutation-symmetric function that takes the coordinates (φ, λ, h) of the OV's corner points, as defined by projecting laterally L_{buff} to the left and right of the two waypoints delimiting the segment, as per Eq.4 for a plan with n waypoints. Circular OVs at take-off and landing are defined solely by R_{tofl} .

$$\left\{ \begin{aligned} &(\varphi_i, -L_{buff}, \lambda_i, -L_{buff}), (\varphi_i, L_{buff}, \lambda_i, L_{buff}), \\ &(\varphi_{i+1}, -L_{buff}, \lambda_{i+1}, -L_{buff}), (\varphi_{i+1}, L_{buff}, \lambda_{i+1}, L_{buff}) \end{aligned} \right\} \quad (4)$$

Horizontal conformance within ov is given by the Boolean $\mathcal{L}_{ov}$, while vertical conformance is given by $\mathcal{A}_{ov}$. The combined 3D conformance check $\mathcal{C}_{ov}$ is defined in Eq.5.

$$\mathcal{C}_{ov}(\varphi_{UAS}, \lambda_{UAS}, h_{UAS}) = \mathcal{L}_{ov}(\varphi_{UAS}, \lambda_{UAS}) \cap \mathcal{A}_{ov}(h_{UAS}) \quad (5)$$

An operation is considered conformant if at least one OV segment, from a flight plan $\mathcal{F}$, satisfies $\mathcal{C}_{ov}$, as per Eq.6. This definition reflects the fact that once a flight plan is activated, the operation volume is valid over its entire active time window, without enforcing intermediate time checkpoints between waypoints. Consequently, the operation may be located in any authorised segment at a given time, allowing for temporary deviations (e.g., due to conflict avoidance) that exit one segment and re-enter another (either the preceding or subsequent segment) without triggering a irreversible contingency, provided the aircraft remains within at least one permitted OV.



(a) Nominal at t_0

(b) Non-conformance at t_1

(c) Predict non-conformance at t'_0

(d) Contingency triggered at t'_1

Fig. 9. Reactive vs predictive conformance monitoring and corresponding flight status updates. Both cases start from identical initial conditions and run over equivalent time intervals, but differ in monitoring configuration: reactive monitoring uses timeline t in (a) and (b), while predictive monitoring uses timeline t' in (c) and (d).

$$\exists ov \in \mathcal{F}_{\text{UAS}}, \mathcal{C}_{ov}(\varphi_{\text{UAS}}, \lambda_{\text{UAS}}, h_{\text{UAS}}) \Leftrightarrow \text{Conformance} \quad (6)$$

The service supports both reactive (current-time) and predictive conformance checks. Prediction extrapolates the trajectory forward in time, accounting for uncertainties such as GNSS noise, weather effects, and manoeuvre delays. If an operation exceeds its OV, its status changes from *activated* to *non-conformant*. As described in Fig.9, non-conformance can be detected reactively or predicted in advance. In the latter case, a contingency procedure can be triggered before the actual breach occurs.

C. Contingency Management

Contingency management ensures high safety, security, and efficiency in operations. A contingent situation arises from a persistent non-conformance status, as shown in Fig. 9d, or through alerts from the operator or the UTM service provider.

Contingency actions are implemented to ensure safety for the contingent vehicle and surrounding traffic. The contingency management protocol is illustrated in Fig.10 and is initiated by the risk map from the Risk Analysis Assistance service (recall Sec. III-C), which identifies SLZ & ELZ.

Once the contingency manager receives operation-specific information, it short-lists feasible SLZ & ELZ based on the contingency level. The nearest unoccupied option is then selected as the new destination and marked as occupied in the database.

The contingent vehicle is considered as a moving obstacle for the rest of the traffic. It is also assumed that the contingent vehicles were semi or fully controllable and operates a conventional landing due to the diverse source of contingency that may restrict vertical landing. Contingency events, their corresponding

TABLE 1. Parameterisation of contingency events to generate operational perturbations.

Disruption	Effect	Occurrence rate*
Loss of power	Constantly loosing altitude	SPV & HPV: 1.0
Loss of link	Disappear for one time step	SPV & HPV: 1.0
Loss of control	Move to a random direction	SPV: 1.0; HPV: 0.5

* Number of event per flight hour.

operational behaviours, and their arbitrarily selected frequencies, chosen to provoke some contingencies while maintaining realism, are summarised in Table 1. Finally, when the contingent operation lands safely, the capacity of the selected SLZ & ELZ is updated back, and the contingency manager block terminates.

D. Tactical Conflict Resolution

The Tactical Conflict Resolution service prevents mid-air collisions by managing manoeuvres' instructions for avoiding conflicting intruders. It addresses short-term conflict scenarios within a short time frame up to 50 seconds ahead and, in rare cases, responds to safety breaches caused by upstream service failures that compromise safe operations.

In many cases, in-flight conflicts arise when a contingent operation fails to maintain its OV. The affected operation must guide itself to a SLZ & ELZ, often neglecting surrounding traffic due to factors affecting manoeuvrability or situational awareness. This can lead to potential incursions into neighbouring OVs, as illustrated in Fig.11. In this illustrative example, aircraft EHG9 loses altitude and encroaches upon the nominal OV of PHAN10, posing a threat. The Tactical Conflict Resolution service issues instructions to the nominal operation, which is temporarily in an off-nominal state due to external factors, in this case, a contingent operation.

When combined with the Contingency Management service (recall Sec. IV-C), Tactical Conflict Resolution functions in two modes based on the presence of contingencies. If no contingencies are detected in the observation layer, the solver remains deactivated. However, if at least one contingency is detected in the traffic, the solver activates and follows the safety layers shown in Fig.11b, which include:

- **Observation:** Positions of contingent operations are compared with those of surrounding traffic. An operation is considered under observation if it falls within this layer, and the solver acts even if there is no imminent threat.
- **Loss of Well-Clear (LoWC):** This layer signifies a loss of separation but with relative severity. The solver acts to resolve conflicts at this level.
- **Near Mid-Air Collision (NMAC):** This represents the highest severity level, where there is a significant risk of collision.

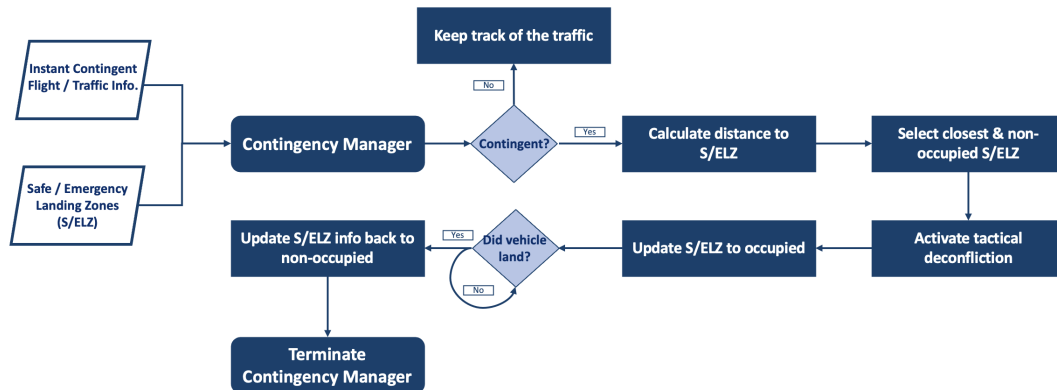
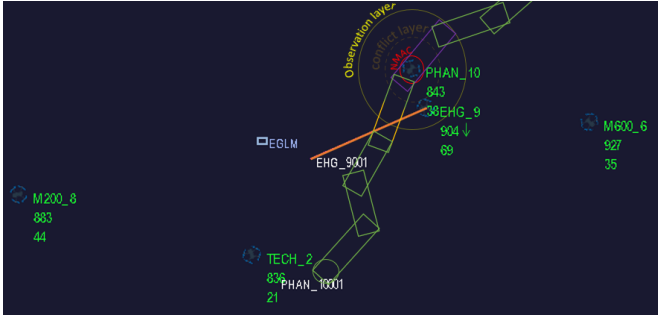


Fig. 10. Contingency Manager Workflow.



(a) Operation volume incursion with flight status



(b) Operation volume incursion with safety layers

Fig. 11. Illustration of a tactical conflict scenario from two perspectives. (a) shows a 3D view where a contingent operation (orange) crosses another OV, resulting in a conflict with the nominal operation (blue). (b) depicts a similar scenario in 2D, highlighting the concentric safety layers (i.e., observation, LoWC, and NMAC) used to trigger Tactical Conflict Resolution.

The solver is trained to minimise intrusion within this layer to prevent potential mid-air collisions.

When the Conformance Monitoring or Contingency Management services are not provided (as we do in Sec.V-C3), the Tactical Conflict Resolution service stays alert. It considers any operation within the activation range (i.e., observation) to be an intruder, which triggers the service's mode to switch from dormant to active.

The service centralises traffic information into an observation vector containing details about the threatened operation and encoded encounter data from both threatened and contingent operations, including parameters such as separation distance, crossing angle, and relative speed. As illustrated in Fig.12, this information is processed using a Long Short-Term Memory network, enabling sequential handling of multiple contingent operations threatening a single aircraft. The resulting observation vector is fed into a shared policy that generates deconfliction instructions, combining altitude manoeuvres in $\{-50, 0, 50\}$ ft

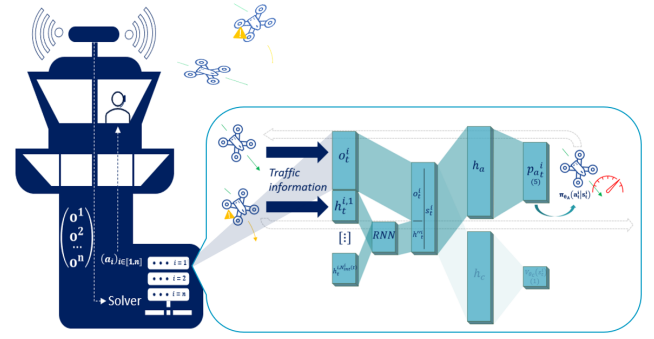


Fig. 12. Conceptual centralised Tactical Conflict Resolution.

with speed adjustments in $\{-5, 0, 5\}$ kts. This shared policy forms the core of the Tactical Conflict Resolution decision-making process, implemented via a Multi-Agent Reinforcement Learning (MARL) system that treats threatened operations as agents requiring deconfliction [11]. While MARL is often recognised for its ability to resolve multi-threat scenarios, it is equally valuable, and often more practical for solving multiple isolated conflicts within seconds through a single centralised entity.

V. EXPERIMENTS

This section presents the experimental studies conducted to demonstrate the proposed advanced UTM services. It begins with the evaluation metrics in Sec.V-A, followed by the experimental setup in Sec.V-B. The subsequent sections report the results: Sec.V-C presents the progressive combination of UTM services and their full capabilities, while Sec.V-D focuses on a sensitivity analysis assessing the influence of different control variables across services. The complete dataset for these experiments is available in an online repository [49].

A. Evaluation Metrics

This section presents the key metrics for evaluating the efficiency of UTM services, categorised into four Key Performance Areas (KPA): Safety, Path efficiency, and workload. Safety is the top priority in research deployment. While this is time-based, the Tactical Conflict Resolution service emphasises in-flight separation and its distribution. Following the ASTM guidelines for UAV Detect and Avoid Systems, the Loss of Well-Clear and Near Mid-air Tolerances are established at 2,000 ft and 500 ft, respectively.

Pre-flight services also focus on path efficiency in flight plan design. Delay measurements in minutes are evaluated through rerouting and rescheduling, guided by the Operation Plan Preparation & Optimisation, Strategic Conflict Resolution, and Dynamic Capacity Management services. Additionally, Risk Analysis Assistance affects cost efficiency with geofence restrictions that necessitate rerouting.

This study also introduces conceptual metrics to evaluate the potential workload associated with operations, which are categorised by flight status that reflects the frequency of vigilance required by the UTM service provider. The process of changing status and in-flight interactions contribute to this workload,

primarily described in the Conformance Monitoring service, in conjunction with the Contingency Management service.

Drawing inspiration from EUROCONTROL's Complexity Metrics for ANSPs [50], this research attempts to adapt them to UTM as follows.

- **Interactions:** The accumulated number of simultaneous operation in the airspace is defined through unidirectional comparisons at each timestep.
- **Horizontal Different Interacting Flow (HDIF):** The interactions per flight hour for operations at the same flight level within a 5NM range.
- **Vertical Different Interacting Flows (VDIF):** The interactions per flight hour during operations transitioning to or from the same flight level with a vertical separation of under 160ft (A_{buff}) and within 5NM horizontally.
- **Speed Different Interacting Flows (SDIF):** The interactions per flight hour for operations with a speed difference greater than 15 kts within a 5NM range.
- **Adjusted Density:** The traffic over a specific period, indicating the ratio of interaction hours to flight hours.
- **Concentration:** The distribution of traffic within the airspace using the ratio of adjusted to raw densities.
- **Structural Index:** The traffic flow structure as the sum of the relative HDIF, VDIF, and SDIF indicators.
- **Complexity Score:** The complexity of operational environment by means of multiplying the Adjusted Density with the Structural Index.

B. Setup

All the case studies in this experiment are based on a synthetic, large-scale scenario representing a portion of the airspace managed by a UTM service provider, which we refer to as our research deployment.

This airspace, as shown in Fig.13, covers 115 NM² with dimension of 12.5 NM by 9.2 NM, and incorporates four discrete Flight Levels (FLs) at 657, 821, 985, and 1,150 ft, which are the nominal altitudes of each level. Each operation is pre-assigned one flight level by the pre-flight services, with HPVs allocated to the highest level and SPVs distributed across the remaining three to balance demand. The planned cruise altitude corresponds to the nominal altitude of the assigned level, with a control tolerance of ± 80 ft for tracking. This defines a vertical band of $[z_i - 80, z_i + 80]$, giving $A_{buff} = 160$ ft. Vehicles do not cruise between levels, except short climb/descend segments occur only to change from z_i to z_j when commanded by take-off, landing, re-routing, and Tactical Conflict Resolution.

Upon receiving flight plans from pre-flight services, a random vehicle model is assigned to each operation based on the vehicle performance (HPV or SPV) outlined in Table 2. To assess the Contingency Management service, synthetic failure events have been created, as detailed in Table 1. The Dynamic Capacity Management service employs a grid-based partition to calculate the cost of each airspace cell shown in Fig. 13. Strategic Conflict Resolution sets the separation minima to 15 minutes. The Tactical Conflict Resolution service uses a shared policy trained through a Multi-Agent Reinforcement Learning system based on Model 7.0 from previous work [11].

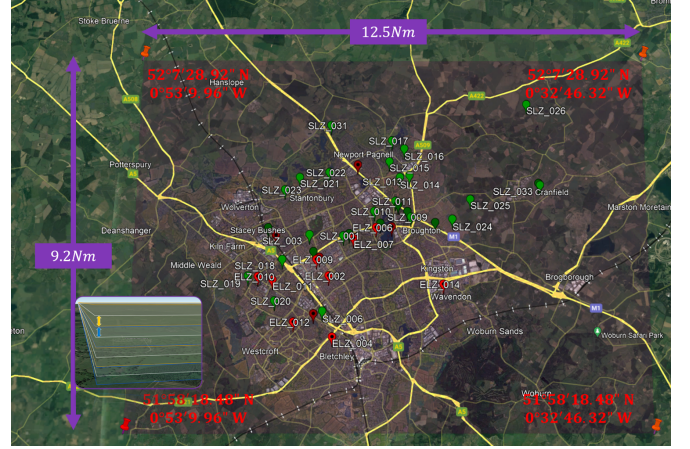


Fig. 13. Airspace Configuration.

TABLE 2. Vehicle performance & operation volume size.

Aircraft model	Perf.	R_{toffl} (m)	L_{buff} (m)	A_{buff} (ft)
Ehang 216	HPV	30	± 250	± 80
Tecnalia		17	± 125	
Twin Mapper 2		15	± 50	
Horsefly Gen 5.3		14	± 125	
DJI Matrice 100	SPV	13	± 125	
DJI Matrice 200		14	± 75	
DJI Matrice 600		13	± 25	
DJI Mavic pro		14	± 25	
Matternet quadrocopter		13	± 25	
DJI Phantom 4		14	± 50	

C. Results - Combination of UTM Services

This section presents the experimental evaluation of different combinations of UTM services, as summarised in Table 3. Three categories of pre-flight service combinations are introduced, corresponding to three sets of flight plans: Initial, Intermediate, and Optimal (noted “ini”, “Inter”, and “OPT”. in the table).

First, the initial set considers only OPP, then the intermediate set adds RAA and DCM to the flight authorization. Finally, the optimal set completes the authorization process with ScR.

We index scenarios as P.I., where P encodes the pre-flight service bundle and I encodes the in-flight bundle. A value of 0 indicates that the corresponding bundle is disabled. For example, 0.0 is our benchmark (no services).

In this set of experiments, each configuration detailed in Table 3 is evaluated under three traffic densities: low (50 operations), moderate (100 operations), and high (150 operations). One-fifth of the operations are HPVs, which operate at the highest FL (i.e., 1,150 ft), while the remaining operations consist of SPVs distributed across the three lower flight levels.

The structure of this section is summarised as follows. First, Sec.V-C1 focuses on comparing the quality of flight authorisations involving different pre-flight services, considering both safety and flight efficiency aspects. Then, Sec.V-C2 examines the safety implications of these flight authorisations at the tactical level. Sec.V-C3 investigates the performances of in-flight services in the absence of most of pre-flight services. Finally, Sec.V-C4 presents the results when all UTM services are jointly activated.

TABLE 3. Combination of advanced UTM services in the experiments.

Flight Plan	Scenario	OPP & OPO ¹	RAA ²	DCM ³	ScR ⁴	MO ⁵	CM ⁶	TcR ⁷
ini	0.0	OPP						
ini	0.1	OPP						✓
ini	0.2	OPP				✓	✓	
ini	0.3	OPP				✓	✓	✓
Inter	1.0	✓	✓	✓				
Inter	2.0	✓	✓		✓			
OPT	3.0	✓	✓	✓	✓			
OPT	All	✓	✓	✓	✓	✓	✓	✓

¹ (OPP & OPO) Operation Plan Preparation and Operation Plan Optimisation;

² (RAA) Risk Analysis Assistance;

³ (DCM) Dynamic Capacity Management;

⁴ (ScR) Strategic Conflict Resolution;

⁵ (MO) Conformance Monitoring;

⁶ (CM) Contingency Management;

⁷ (TcR) Tactical Conflict Resolution.

1) Initial vs. Optimal Flight Planning : In this section, we compare the initial (“ini”) and optimised (“OPT”) flight plans, as detailed in Table 3. Key performance metrics for these operations, across low, moderate, and high traffic densities, are then summarised in Table 4.

We begin by evaluating the effectiveness of the RAA service based on the order of services applied. In three traffic scenarios, 26, 56, and 83 operations from the initial flight plans were identified as passing through high-risk areas, comprising over half of the operations in each scenario. The total high-risk feedback suggests that some operations have only one segment with unacceptable risk, while others have multiple high-risk segments. Compared to the optimised results, all high-risk operation segments have been successfully cleared through rerouting.

Rerouting issued by the DCM service reflects how many operations have been affected with alternative routes to bypass hotspot areas. As mentioned earlier, the optimisation model calculates the new route in identifying the lowest-cost solution. Table 4 shows that in a low traffic density scenario, a unique operation was rerouted, while in a high traffic density scenario, 7 operations were rerouted. Furthermore, the average route

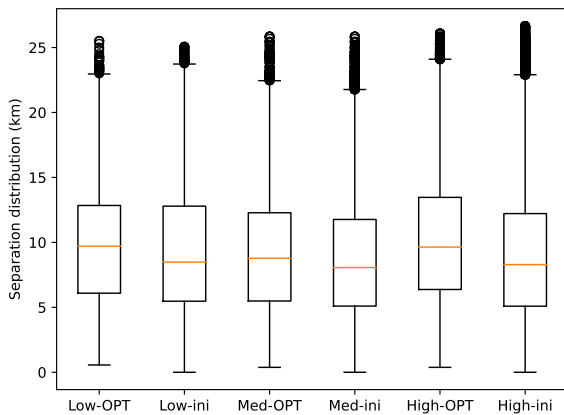
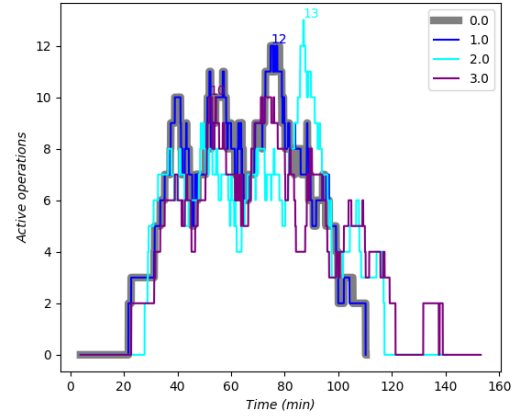


Fig. 14. Separation distributions over various traffic densities.

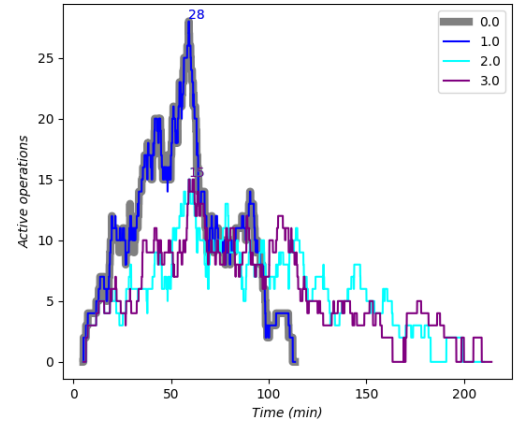
distance may be affected by the RAA and DCM services, as operations can be rerouted after evaluation by these services.

Analysing all submitted flight plans, we found 21, 61, and 101 conflicting operations in the initial experiments without the ScR service. All conflicts can be effectively resolved using this service, as indicated in the optimised result columns.

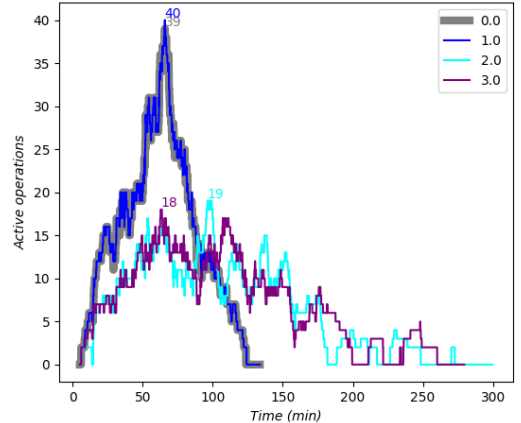
2) Flight Planning Impacts at the Tactical Level : Based on the pre-flight service configurations that generated the initial (“ini”) and optimised (“OPT”) flight plans, this section examines



(a) Low traffic density



(b) Moderate traffic density



(c) High traffic density

Fig. 15. Pre-Flight services effects on airspace occupancy.

TABLE 4. Comparison of flight planning performances with (“OPT”) and without (“ini”) provision of pre-flight services.

Key Metric		Low density		Moderate density		High density	
		ini	OPT	ini	OPT	ini	OPT
Total delay	(min)	0	455	0	3,320	0	5,565
Max delay	(min)	0	90	0	190	0	200
Delayed operations	(#)	0	19	0	58	0	93
Number of conflict	(#)	21	0	61	0	101	0
Rerouting issued by DCM	(#)	0	1	0	2	0	7
High-risk area incursions	(#)	26	0	56	0	83	0
High-risk feedback by RAA	(#)	36	0	93	0	136	0
Route distance for SPV	(km)	4.55	4.55	4.66	4.67	4.73	4.74
Route distance for HPV	(km)	12.05	12.05	13.17	13.17	13.95	13.96

their effects during real-time, simulated operations. Fig.15 shows how the number of simultaneous operations evolves under our traffic scenarios. “ini” corresponds to benchmark Scenario 0.0, while “OPT” is Scenario 3.0. The intermediate configurations (“Inter”) are Scenarios 1.0 and 2.0 (refer to Table 3).

Consistent with the delay statistics in Table 4, the addition of ScR (in Scenarios 2.0 and 3.0) produces larger, but more evenly distributed delays in the moderate and high density cases. In contrast, DCM’s effect alone on flight efficiency is minimal, evidenced by the near overlap of the cyan and magenta curves in Fig.15. DCM stands out only under low-density traffic: while ScR also introduces some delay in this case, the effect is modest. DCM yields the lowest sustained number of simultaneous operations but exhibits the highest peak (13 aircraft around 90 minutes).

As expected, the absence of ScR results in intense traffic in shorter scenarios, as visible for the “ini” configuration and Scenario 1.0 in blue. Fig.14 illustrates the distribution of in-flight separations between pair-wise operations across various traffic: “Low”, “Med”, and “High” for the respective densities. These separations are determined for pairs operating simultaneously at the same altitude. Since all conflicts are resolved by pre-flight services in the optimised flight plan, the minimum separations in all optimised groups exceed those in the initial group, thereby meeting the safety separation threshold. Additionally, the average separation in all three scenarios has increased after processing by the pre-flight services, with the high traffic density scenario showing a more significant improvement. Although the overall comparison shows high similarity, the “ini” configuration shows the most hazardous separations, visible with its first quartile that lies closest to zero.

3) Intervention of Tactical Services : The previous section presented simulated operations under the three pre-flight configurations, which are “ini,” “Inter,” and “OPT”, assuming nominal conditions without any real-time monitoring. In this section, we explore how tactical (in-flight) services can enhance and adapt operational performance relative to those pre-flight plans through three experiments.

In the first experiment, we evaluate the safety benefits of TcR on flight plans that have not been deconflicted (i.e., no ScR). Fig.16 compares our benchmark Scenario 0.0 (no in-flight services) with Scenario 0.1, in which TcR engages whenever horizontal separation drops below 6,000 ft or vertical separation below 100 ft.

In the low-density case illustrated in Fig.16a, both curves show some loss of separation, but TcR notably increases the

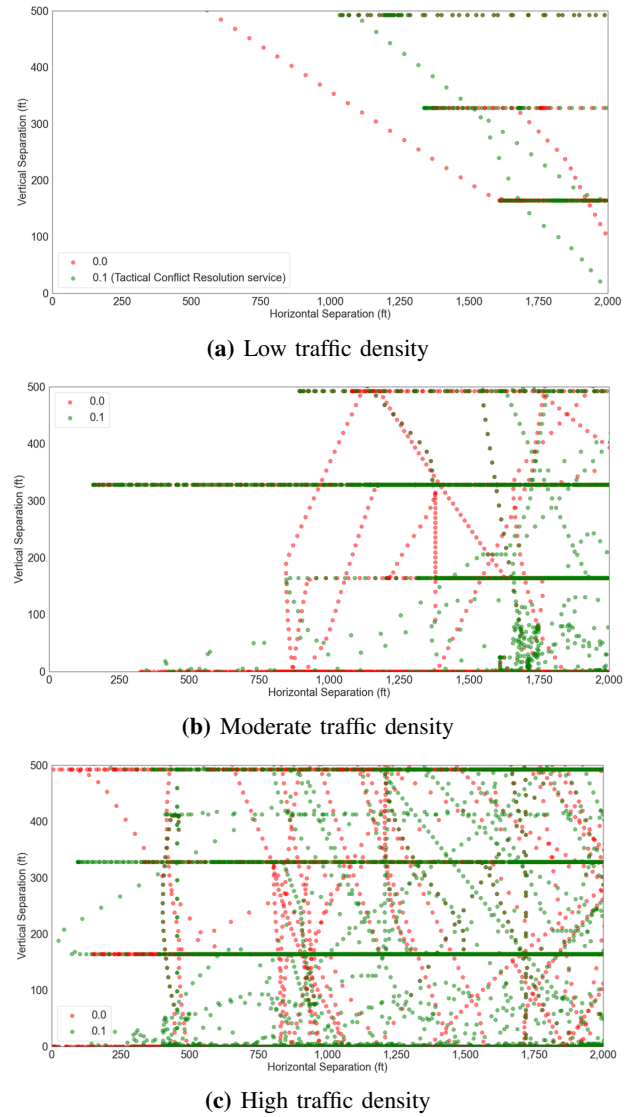


Fig. 16. Pair-wise separation clusters for the “ini” flight plan under three traffic densities (Scenario 0.1), illustrating the safety impact of TcR. Red markers show separations without TcR, whereas green markers show separations with TcR fully active. Points in the lower-left region indicate the most hazardous (closest) encounters.

minimum horizontal spacing between the closest aircraft. Under moderate density (Fig.16b), TcR “pushes back” operations even further apart horizontally, reducing collision risk. At high density (Fig.16c), TcR alone cannot fully prevent NMAC events (horizontal separation <500 ft), though its vertical separation adjustments still avert some overlaps that occur when TcR is inactive.

The second experiment examines the synergy between Conformance Monitoring and Contingency Management services through Scenario 0.2. Its goal is to demonstrate how Conformance Monitoring can spot off-nominal operations and how Contingency Management can respond to these situations. We primarily focus on the workload aspects of the operations as a service provider point of view. Contingency events are based on

TABLE 5. Operation flight status distribution and interactions based on “ini” flight planning (Scenario 0.2)

	Low	Moderate	High
Nominal	99.38%	98.92%	97.07%
Non-conformance off-nominal	0.00%	0.00%	0.01%
Contingent off-nominal	0.62%	1.08%	2.92%
Flight status switch	3	18	69
Interactions (Notifications)	358	3746	9030

TABLE 6. Operation flight status distribution and highest safety critical situation with in-flight monitoring based on “ini” flight planning (Scenario 0.3).

	Distribution (%)			Minimum separation (ft)			Pair Status*		
	Low	Moderate	High	Low	Moderate	High	Low	Moderate	High
N ¹	98.81	96.39	98.57	647	325	7	N-N	N-N	N-N
NC ²	0.02	<0.01	0.02	>3,000	>1,800	>1,800	NC-N	NC-C	NC-C
C ³	1.17	3.59	1.40	>3,000	>900	>900	C-N	C-N	C-N
S ⁴	0.00	<0.01	0.01	-	>1,800	>1,800	-	S-C	S-C

* Status of the closest pair of operation (i.e., having minimum separation).

¹ Nominal flight status;

² Non-conformant flight status;

³ Contingent flight status;

⁴ Nominal flight status in conflict.

the failure rates and protocols listed in Table 5. An interaction is considered to involve a pair of operations if at least one is under off-nominal conditions. This indicates that arbitrary failure rates yield a contingency occurrence rate of up to 3% in the busiest scenario. While this does not reflect real-world UAM expectations, it likely overestimates for testing purposes.

Table 5 quantifies this workload across different traffic densities, highlighting how potential workload escalates with higher density due to increased interactions and risk of separation loss. Conformance Monitoring manages flight status to reduce workload and enhance safety through a set of control variables related to OV, including size and sensitivity of non-conformance activation. In Scenario 0.0, without any service, the UTM service provider must closely monitor operations of 243, 972, and 2,345 thousand flight interactions for the respective densities. According to the same table, Conformance Monitoring reduces this by 679, 259, and 260 times, respectively. These results are sensitively influenced by the frequency of off-nominal situations.

The framework in Sec.II reminds that the activation of the in-flight services are relying on the monitored flight status of the operation considered by the service. For the third experiment, Table 6 summarises the distribution of the different flight status met for the different densities of traffic and the most critical situation for each: the minimum separation observed between a pair of operations and their respective flight status.

Off-nominal operations (i.e., non-conformance or contingency events) remain anecdotal compared to nominal operations, which account for nearly 99% of the operational lifecycle. While off-nominal cases exhibit relatively safe separations, with no NMAs and only a few LoWCs, nominal operations reveal hazardous separation minima, reaching as low as 647 ft, 325 ft, and even 7 ft, as shown in Fig.16.

4) *Unified UTM Services* : We evaluate the integrated system, with all in-flight services running concurrently alongside OPT-based flight planning. Experiments are restricted to moderate traffic density to focus on safety outcomes. Fig.17 compares

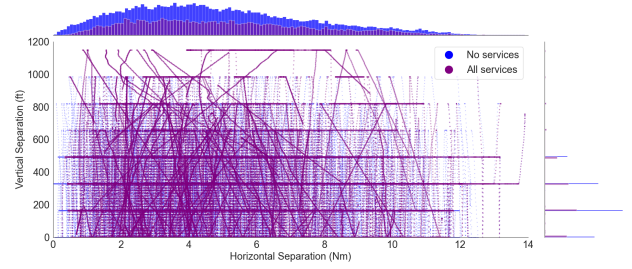


Fig. 17. Horizontal and vertical separation distributions for all flight operation pairs with no services (i.e., blue distribution) and with all in-flight services enabled (i.e., purple distribution). Top and right histograms show marginal separation distributions; the main scatter plot depicts pairwise separation values, with lines indicating clustered vertical levels.

separation distributions for the baseline with no services (i.e., Scenario 0.0) versus all services enabled (i.e. “All”). Fig.18 further stratifies separations by flight operation-pair flight status and clusters them.

First, we observe a clear reduction in simultaneous operations when all services are enabled relative to the baseline with no services. Across all horizontal separations the purple distribution is markedly lower, while both conditions retain a similar peak at 2 to 4NM, encounters below 0.3NM become nearly absent. Vertically, separations concentrate at 200 and 300ft with no material change in proportions, suggesting that flight planning was not significantly altered in the flight level assignments by pre-flight services or, at least, was fairly distributed.

In this experiment, nominal operations account for approximately 85% of total flight time, while off-nominal conditions exceed 15%. Contingent flights remain well separated from surrounding traffic, with the minimum horizontal separation is

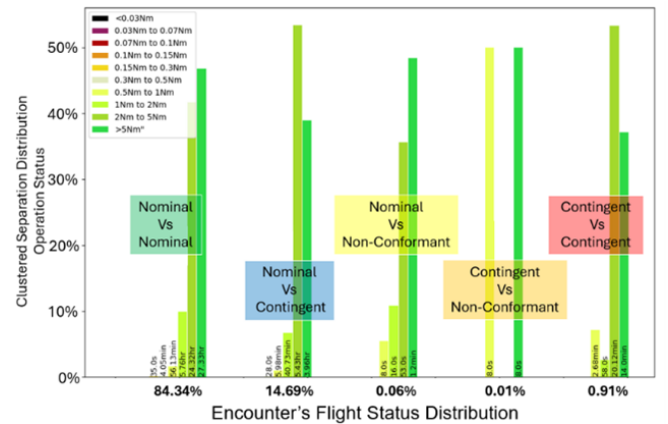


Fig. 18. Clustered horizontal separations by encounter flight-status pairings. Percentages on the horizontal axis indicate each pairing’s share of total encounter time, while colours show separation bins, with a green-to-black gradient indicating increasing severity.

TABLE 7. Complexity metrics for a moderate density airspace, comparing no services (i.e., benchmark Scenario 0.0) versus all UTM services enabled (i.e., Scenario ‘All’). Metrics are reminded in Sec.V-A.

UTM Services	Raw Density a/c.NM ⁻²	Flight hours	Hours of Interactions	Adjusted Density	Concentration	HDIF	VDIF	SDIF	Structural Index	Complexity score
None	0.87	19.57	134.99	6.90	7.93	0.86	0.17	0.07	1.10	7.58
All		19.59	69.28	3.54	4.07	0.32	0.13	0.04	0.48	1.71

above 0.5 NM. As a result, no TcR activations were required, even across repeated runs. Non-conformance occurs only occasionally (less than two minutes over tens of flight hours) and maintains safe separations. Consistent with Table 6, encounters between contingent flights are the most critical, as trajectory changes may be infeasible. By contrast, OPT-based planning shows substantially larger safety margins.

Using the ATM-derived complexity metrics adapted to UTM, enabling all UTM services reduces overall complexity by 4.4 times (Table 7). Hours of interactions, adjusted density, and concentration are each roughly halved. HDIF drops by 63%, while speed and vertical differentials decrease more modestly. The structural index falls by 56%.

D. Results - Sensitivity Analysis

We conducted sensitivity experiments to quantify how key control variables affect service performance: Geofence (obstacle) radius in Sec.V-D1, separation minima in Sec.V-D2, buffer sizing and conformance look-ahead time in Sec.V-D3, and TcR activation range together with progressive operational failure rate in Sec.V-D4.

1) *Geo-fence Sizing* : The size and location of the geofence around potential obstacles provide feedback for the Risk Analysis Assistance service to direct flights away from high-risk areas. The radius is determined by the risk level of the flight segment. For operational efficiency, the geofence is modelled as a cylinder, with the obstacle radius sometimes restricted to prevent major deviations from the flight plan. By keeping the radius within a

certain range, risks are mitigated while minimising unnecessary detours.

The cost-efficiency of the geofence radius on the Risk Analysis Assistance service is shown in Fig.19. It uses the average flight distance in a moderate density configuration to demonstrate the effects of geofence sizing. The impact of the radius on flight distance is expected to follow a similar trend in the other two traffic density scenarios, which is why this paper focuses on moderate traffic density. As mentioned in Sec. III-B, the initial flight plan produced by the Operation Plan Preparation and Optimisation service results in the shortest trajectory distance.

The average distance for both SPVs and HPVs without Risk Analysis Assistance is the smallest among all radius values.

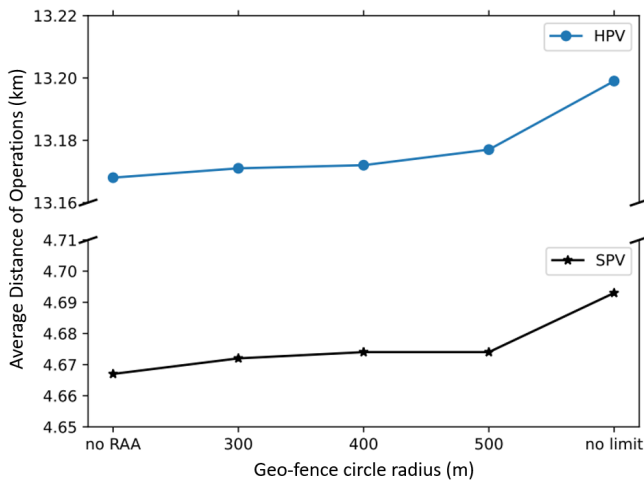
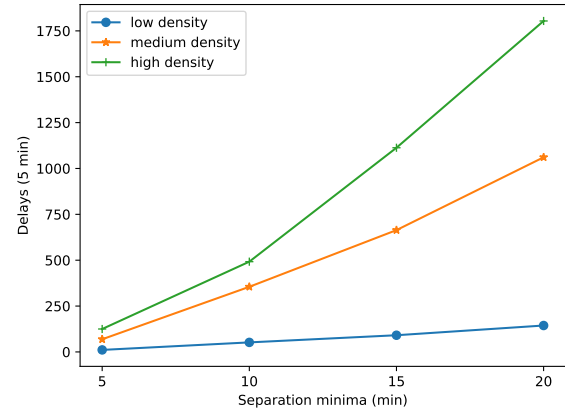
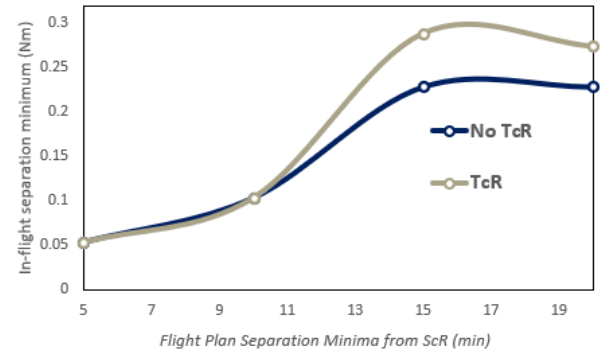


Fig. 19. Impact of geo-fence radius on SPV (i.e., black) & HPV (i.e., blue) routing and operational endurance.



(a) Effects on flight delay



(b) Effects on minimum in-flight separation.

Fig. 20. Impact of Strategic Conflict Resolution separation minima on operations: (a) average flight delay for three traffic densities, and (b) minimum observed in-flight separation, with and without Tactical Conflict Resolution.

Changes in average flight distances are minor when the maximum allowed radius ranges from 300 to 500 meters. Without a radius limit, SPVs and HPVs can operate at maximum distances of 1,390 and 994 meters, respectively, leading to a significant increase in average distances. However, the overall impact of the geofence radius on distance remains relatively minor.

2) *Operation Separation Minima* : As discussed in Sec. III-E, time-based separation is used to identify conflicts between operation pairs in the same airspace volume. Therefore, the separation minima significantly affect the Strategic Conflict Management service regarding delay allocation and conflict resolution. Additionally, safety separation may influence subsequent Tactical

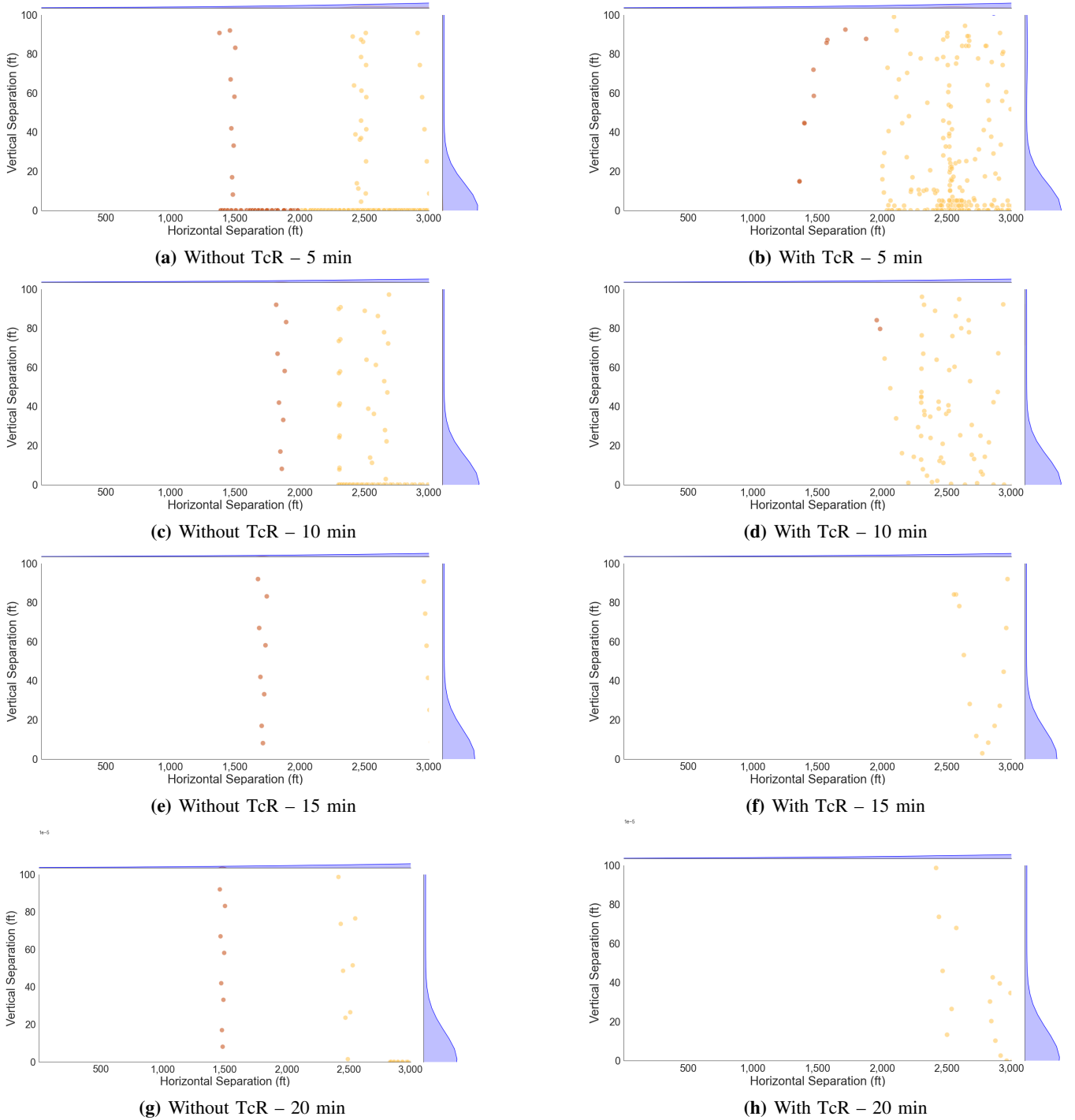


Fig. 21. Safety impact of separation minima, issued by the Strategic Conflict Resolution, on in-flight separations. Pairwise separations for all active aircraft encounters in a moderate density scenario under varying strategic separation intervals (5, 10, 15, and 20 minutes). Left-side figures (a), (c), (e), and (g) present the results without Tactical Conflict Resolution, while the corresponding right-side figures (b), (d), (f), and (h) show the results with active Tactical Conflict Resolution.

Conflict Resolution services. To explore the relationship between these two services, we conducted sensitivity experiments varying safety separation minima while keeping other factors constant.

We first examine how separation minima affect Strategic Conflict Resolution services under various traffic scenarios. As shown in Fig. 20a, increasing the safety separation minima leads to longer delays, particularly at higher traffic densities. Typically, a 5-minute increase in separation minima results in a delay increase of 1.5 to 2 times. Transitioning from a 5-minute to a 10-minute safety interval causes a more pronounced delay increase, as a 5-minute interval allows most flights to operate conflict-free, minimising delays.

We operate various flight plans based on four separation minima: 5, 10, 15, and 20 minutes, to evaluate the behaviour of the Tactical Conflict Resolution service. Fig. 21 compares the four scenarios with (TCR) and without (No TCR) the service, displaying separation distribution. The Tactical Conflict Resolution service enhances in-flight separation similarly to Fig. 14, even when flight plans are already deconflicted. It eliminates separation below 2,000 feet when employing a 10-minute minimum or greater. The impact of a 5-minute separation minimum is more contentious, as it slightly increases vertical separation beyond 1,500 feet, while the initial scenario indicates horizontal separation just below 1,500 feet. This analysis is illustrated in Fig. 20b, which compares in-flight separation with the minima set by the Strategic Conflict Resolution service.

3) *Operation Volume Buffer Sizing* : This analysis emphasises the safety implications of the OV's buffer size, particularly in managing non-conformance sensitivity. A smaller OV improves situational awareness by being more responsive to deviations, allowing for timely decision-making during off-nominal conditions. However, while a smaller volume with a larger separation minima enhances responsiveness and safety, it may lead to undesirable non-conformance statuses if the buffer size does not align with the flight envelope. Fig. 22 illustrates the proportion of undesirable non-conformance statuses as prediction time varies from 5 seconds to 2 minutes.

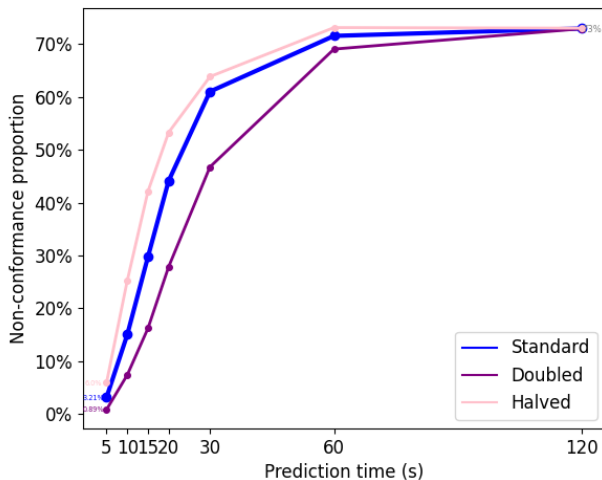


Fig. 22. Effects of buffer size on conformance prediction.

Prediction time for conformance significantly influences the frequency of false non-conformance alerts. All three buffer size configurations identify non-conformance in about 73% of operations. The remaining 27% occur during stable in-cruise phases, where the trajectory remains unchanged. While predicting non-conformance based solely on time-ahead calculations with current navigational data is straightforward and safe, it results in a high rate of false alarms, making it less effective. Even with a doubled buffer size, nearly 1% of operations still face these alerts.

4) *Dependency Against Contingencies* : This analysis examines the resilience of Contingency Management and Tactical Conflict Resolution services in chaotic environments. It builds on the findings in Sec.V-C, highlighting limitations in interpreting the benefits of Tactical Conflict Resolution for mitigating contingent encounters. According to the standard failure rates, the failure rate has increased by up to four times. Fig. 23 depicts the separation distribution between pairs of off-nominal operations. Specifically, Fig. 23a focuses on interactions between contingent aircraft, while Fig. 23b examines conflicts involving operations that observe surrounding contingency scenarios. These are analysed based on activation ranges set between 0.3 NM and 1 NM.

No evident correlation is observed between the number of simultaneous contingent operations (i.e., interactions) and failure rates. This sensitivity analysis is constrained by the randomness of failure events, which can be rapidly mitigated when operations

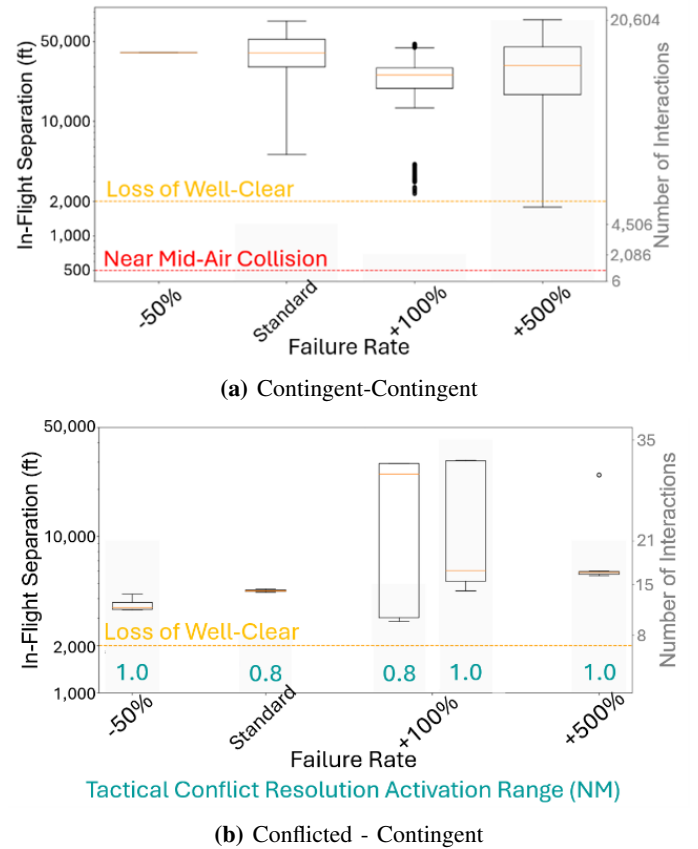


Fig. 23. Pairwise in-flight separations under escalating contingency events and varying tactical conflict detection sensitivity.

are near SLZ & ELZ, even as the number of contingencies may increase. This is evident in Fig. 23a, where doubling the failure rate results in fewer interactions between pairs of contingent operations. However, a fourfold increase in the standard failure rates leads to chaotic scenarios, with a significant proportion of traffic becoming contingent. Despite these challenging circumstances, separations generally remain safe, with only a few instances of operations breaching the well-clear tolerance, which did not escalate into more severe issues.

Despite the chaotic conditions observed in high failure rate scenarios, the Tactical Conflict Resolution service remains largely inactive, irrespective of the activation range settings. Specifically, across multiple high traffic density scenarios, no tactical conflict resolution was triggered when the activation range was set strictly below 0.8 NM. As illustrated in Fig. 23b, interactions between conflicted operations managed by the service and contingent operations occurred over brief intervals without any loss of separation. The broad separation distribution observed under a doubled failure rate highlights cases where separations exceed the activation range, with some operations maintaining distances as large as 10,000 ft.

VI. DISCUSSION

This section is organised as follows: Sec.VI-A presents the experimental outcomes along with a cross-analysis of the results. Sec.VI-B highlights the main limitations and challenges, and Sec.VI-C discusses directions for future work.

A. Key Outcomes

Across all experiments, total delay and the number of affected operations grow with traffic density as depicted in Table 4. Under a FCFS regime, early delay allocations propagate to subsequent flights and amplify both average and maximum delays, especially in the moderate and high-density cases. Sensitivity to separation minima reinforces this: increasing the strategic threshold raises delay non-linearly as illustrated in Fig.20 where separation minima from 5 min to 10 min step often causing a marked jump. These results highlight the importance of coupling strategic deconfliction with policies that minimise knock-on effects in congested periods.

Conformance Monitoring and Contingency Management are central to safe operations but also drive provider workload: every status change and interaction is a potential distraction or triggers downstream action as detailed in Table 5. The effectiveness of Conformance Monitoring depends on timely, trustworthy inputs from both pre-flight authorisations and in-flight state estimates. If a non-deconflicted plan bypassed Strategic Conflict Resolution, an incipient hazard could be masked and go unmitigated. Our buffer-sizing analysis in Sec.V-D3 further shows that low time-ahead conformance prediction can lead to high false-alert rates. To mitigate alert fatigue, buffer size and look-ahead should be tuned to the flight envelope, and ideally, trajectory-aware predictors should be employed.

As shown in Fig.17 when all UTM services are enabled, we observe a clear reduction in simultaneous operations and

a healthier separation profile relative to the baseline. The distributions highlighted in the results suggest that pre-flight services did not materially distort flight-level allocations. In the same experiment, among nominal–nominal encounters the vast majority exceed 1 NM, although short-lived passes at 0.15 and up to 0.3 NM (about 35s) appear across adjacent flight levels and would be benign unless an off-nominal event arose mid-encounter. Encounters involving contingencies remain mostly above 0.5 NM, and repeated runs required no Tactical Conflict Resolution activations at the tested settings, consistent with Strategic Conflict Resolution providing adequate strategic margin (Fig.18, Table 6).

These local separation gains translate to macro-level simplicity: using ATM-derived metrics adapted to UTM, enabling all services reduces overall complexity as detailed in Table 7. Raw density and flight hours remain essentially unchanged, isolating the effect to service enablement rather than scenario scaling. In scenarios without Strategic Conflict Resolution, Tactical conflict Resolution increases closest horizontal spacing and mitigates risk across densities but cannot fully offset congestion at high density according to Fig.16, recommending upstream strategic deconfliction.

The design levers behave as expected. Geofence sizing has a modest average impact on route distance and endurance (Fig.19). Limiting the radius in low-risk areas helps avoid unnecessary detours, whereas allowing larger radii around high-risk obstacles is acceptable provided that endurance margins are maintained. Strategic separation minima present a classic safety–cost trade-off: larger thresholds raise delays but, together with TcR, eliminate short-range encounters below 2,000 ft at 10 min or greater (Fig.20, Fig.21). The placement and capacity of SLZ & ELZ also matter in practice: while contingent–contingent separations remained safe in our runs, simultaneous nearby contingencies could exceed system capacity without further safeguards.

B. Limitations

This study has several limitations. Results are derived from synthetic scenarios with a single airspace geometry and a simplified traffic mix, which may not capture city-specific constraints. The simulation stack is deliberately simple, idealised vehicle kinematics, perfect communications/sensing, no weather or GNSS degradation, and limited state-estimation error, which can understate corner cases. Failure rates and contingency behaviours are assumed rather than empirically calibrated. Fourth, pre-flight and in-flight services are only loosely coupled: there is no replanning loop, shared state interface, or latency/quality-of-service contracts ensuring tight coordination within the time window between flight plan approval and operation activation. Human-in-the-loop evaluation is absent, so impacts on UTM providers' workload, trust, and acceptability are not accurately measured. The framework is not integrated with ATM/ATC systems or real aircraft operations. UTM–ATM data exchange (e.g., constraints, advisories, tactical clearances) and operational collaboration with ATC are not represented. Detailed aircraft-performance models (energy, climb/descent envelopes) are not yet embedded in curvature or level-assignment constraints. These factors limit external validity and policy transferability. Finally,

although the underlying framework includes Digital Twin capabilities, such as live–virtual–constructive integration and real-time mirroring—these features were only partially employed in this study. The current deployment relied primarily on simulation, without exploiting full Digital Twin mechanisms, which limits the extent to which Digital Twin benefits are demonstrated in the results.

C. Future Work

A practical next step is to close the loop by feeding real-time insight back into some pre-flight services during the time window between flight plan approval and operation activation. This would allow dynamic updates to Operation Plan Preparation & Optimisation with temporary restrictions, refresh risk maps as SLZ&ELZ occupancy changes, and pass predictive conic envelopes of contingent trajectories to Strategic Conflict Resolution to pre-empt conflicts.

We also plan more advanced simulations to stress services under richer interactions such as mixed vehicle classes, interacting contingencies, stochastic failures and to run human-in-the-loop studies. These will help identify which services can be safely automated and those requiring human supervision, and quantify effects on workload, trust, and performance.

Beyond this, we aim to tighten collaboration between pre-flight and in-flight modules via shared APIs and latency budgets, enabling real-time capacity updates and sustainable coordination in an integrated ATM/UTM R&D testbed. We will broaden scope to include additional operational features (e.g., noise or privacy constraints) and scaling to city-wide, federated providers.

Finally, we intend to establish open, reproducible benchmarks and KPIs for UTM services, together with datasets and reference implementations in light of latest progress, to support comparative research and inform stakeholders. Parallel efforts will incorporate detailed aircraft-performance models into curvature constraints and validate findings with real geofences and operational data.

VII. CONCLUSION

This study introduced a co-simulation environment that integrates a comprehensive suite of collaborative UTM services, enabling realistic and scalable evaluation of their performance. Across more than 1,600 simulated flight hours, the framework demonstrated the ability of advanced UTM services to manage high-density urban airspace, handle complex contingencies, and maintain operational safety. The results highlight both the strengths and the boundaries of these services, showing the need for continued integration research, adaptive decision-making algorithms, and robust benchmarking methods to support the transition from conceptual designs to deployable UTM systems.

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VIII. BIOGRAPHY

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